

Dual IPv4 and IPv6 UDP with NetX Duo on NUCLEO-H743

Status of the STM32CubeIDE firmware example as of 2 September 2026. UDP server on port 6000, LED remote control, button reporting, serial console, documentation.

Date	2 September 2026	Release	1.0.0 (31 August 2026)
Period	31 August 2026 to 2 September 2026	Status	Baseline complete. Lab bind pending.
Board	NUCLEO-H743ZI2, STM32H743ZIT6, Cortex-M7	Stack	ThreadX 6.2 + NetX Duo 6.2
Tools	CubeMX 6.15.0, X-CUBE-AZRTOS-H7 3.1.0	Repo	ccsalman545/crispy-octo-computing-machine

Complete Firmware 1.0.0: dual-stack UDP, LED and button paths, full docs set.	In lab Static IPv4 and IPv6 must match the local subnet before tests.	Open DHCP, IPv6 button reports, and TCP sockets are out of scope.
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1. Purpose

Record the 1.0.0 baseline on NUCLEO-H743ZI2. ThreadX 6.2 and NetX Duo 6.2 run one UDP socket on port 6000 for IPv4 and IPv6. Text commands drive LD1 (PB0). Button B1 (PC13) is polled with a two-read debounce and reported to a host PC. USART3 (115200 8N1, ST-Link VCP) prints addresses, the bind line, and every payload. CubeMX generates HAL, linker scripts, and startup. IPv6 is enabled by hand in app_netxduo.c because CubeMX does not emit it.

2. Work completed

- UDP server bound to port 6000. Dual-stack receive path is live.
- LED parser: LED ON, LED_ON, LEDON, ON and the matching OFF forms, including case variants.
- Button stream: BUTTON:0 pressed, BUTTON:1 released, to 192.168.1.160:55100 every 20 ticks (~200 ms).
- Addressing: IPv4 192.168.1.111/24, IPv6 2600:1702:4eb2:9780::abcd/64. Link-local from the MAC.
- Ethernet: LAN8742 RMII, MAC 00:80:E1:00:00:00, DMA descriptors at 0x30040000, 128 KB non-cacheable MPU region.
- Pools: 10 x 1536 B packets, ARP 1024 B, 30 KB NetX pool in RAM_D2. Two priority-10 threads, 2 KB stacks, 100 ticks/s. Docs and PR #1 merged 31 August 2026.

3. Runtime path

Reset_Handler, SystemInit, main(): MPU_Config, I/D-cache, HAL_Init (TIM6), SystemClock_Config (HSE bypass + PLL), MX_GPIO_Init, MX_ETH_Init, MX_USART3_UART_Init, MX_ThreadX_Init. tx_kernel_enter does not return. tx_application_define creates the 1 KB ThreadX pool and the 30 KB NetX pool, then MX_NetXDuo_Init: nx_system_initialize, nx_packet_pool_create, nx_ip_create (192.168.1.111 / 255.255.255.0, nx_stm32_eth_driver), ARP/ICMP/TCP/UDP enable, nxd_ipv6_enable, nxd_ipv6_address_set (/64, interface 0), nxd_icmp_enable, tx_thread_create(nx_app_thread_entry). The thread prints addresses, binds UDP 6000, then loops: read PC13, debounce, send BUTTON:n, sleep 20 ticks, receive with a 1-tick timeout, parse LED commands, release the packet.

4. UDP protocol and endpoints

Direction	Address / port	Payload	Effect
PC to board	192.168.1.111 port 6000	LED ON, LED_ON, LEDON, ON	LD1 (PB0) high
PC to board	[2600:1702:4eb2:9780::abcd] port 6000	LED OFF, LED_OFF, LEDOFF, OFF	LD1 (PB0) low
PC to board	Same socket, IPv4 or IPv6	Any other text	Echo on USART3
Board to PC	192.168.1.160 port 55100	BUTTON:0	B1 pressed
Board to PC	192.168.1.160 port 55100	BUTTON:1	B1 released

5. Configuration snapshot

Item	Value	Where
IPv4 / mask	192.168.1.111 / 255.255.255.0	app_netxduo.h lines 87, 89
IPv6 global / prefix	2600:1702:4eb2:9780::abcd / 64	app_netxduo.c lines 265 to 268
UDP listen / report	6000 listen, 55100 report	app_netxduo.c lines 323, 365
MAC / console	00:80:E1:00:00:00, USART3 115200 8N1	main.c lines 229 to 234, 276
Packet pool / stacks	10 x 1536 B, 2 KB stacks, priority 10	app_netxduo.h lines 47, 73, 77 to 83
NetX byte pool	30 KB in .NetXPoolSection (RAM_D2)	app_azure_rtos_config.h
Debounce / PHY	2 reads, 20 ticks; LAN8742 RMII	nx_app_thread_entry; .ioc

6. Verification status

ID	Check	Pass when	State
T1	Console bring-up	Prints IPv4, IPv6, and UDP Server listening on PORT 6000..	Ready to run
T2	IPv4 LED on/off	LED ON / LED OFF (and ON / OFF) drive LD1; console confirms	Ready to run
T3	IPv6 LED on/off	LED_ON / LED_OFF to the /64 address drive LD1	Network dependent
T4	Unknown payload	hello is echoed on USART3; LED unchanged	Ready to run
T5	Button stream	BUTTON:1 idle, BUTTON:0 while B1 held, ~200 ms period	Ready to run
T6	ICMPv4 / ICMPv6	ping 192.168.1.111 replies; ping6 to ::abcd if the LAN routes it	IPv4 ready; IPv6 LAN

Ready to run means the code path and the matrix in docs/TESTING.md exist. Pass or fail is recorded on the target LAN after static addresses match that LAN. IPv6 unicast through consumer routers is not guaranteed. Use the same /64, or test link-local.

7. Open items and risks

Item	Impact	Action
IPv6 address is lab-specific	Unreachable on another /64.	Edit the four v6 words in app_netxduo.c.
IPv4 is static only	Wrong subnet or duplicate 192.168.1.111.	Align NX_APP_DEFAULT_IP_ADDRESS.
Button reports IPv4 only	IPv6-only hosts never see BUTTON:n.	Add an NXD_ADDRESS v6 target if required.
TCP enabled, unused	Uses RAM with no application socket.	Leave as-is, or drop nx_tcp_enable.
Error_Handler is silent	Bind or IPv6 failure looks like a hang.	Use USART3 and docs/DEBUGGING.md.

8. Next steps (in order)

1. Set IPv4, IPv6, MAC, and the PC report address to the lab network. Rebuild and flash via ST-Link.
2. Run T1 to T6. Keep the USART3 transcript and Packet Sender captures for IPv4 and IPv6.
3. Confirm B1 reporting on UDP 55100. Change SEND_SLEEP_TICKS only if the host cannot keep up.
4. Do not add DHCP, TCP sockets, or extra threads until dual-stack UDP is signed off on hardware.

Prepared 2 September 2026 from repository 1.0.0 (changelog 31 August 2026). Sources: NetXDuo/App/app_netxduo.c, app_netxduo.h, Core/Src/main.c, AZURE_RTOS/App/app_azure_rtos.c, docs/ARCHITECTURE.md, docs/TESTING.md, docs/CONFIGURATION.md. No scope change is proposed in this report.